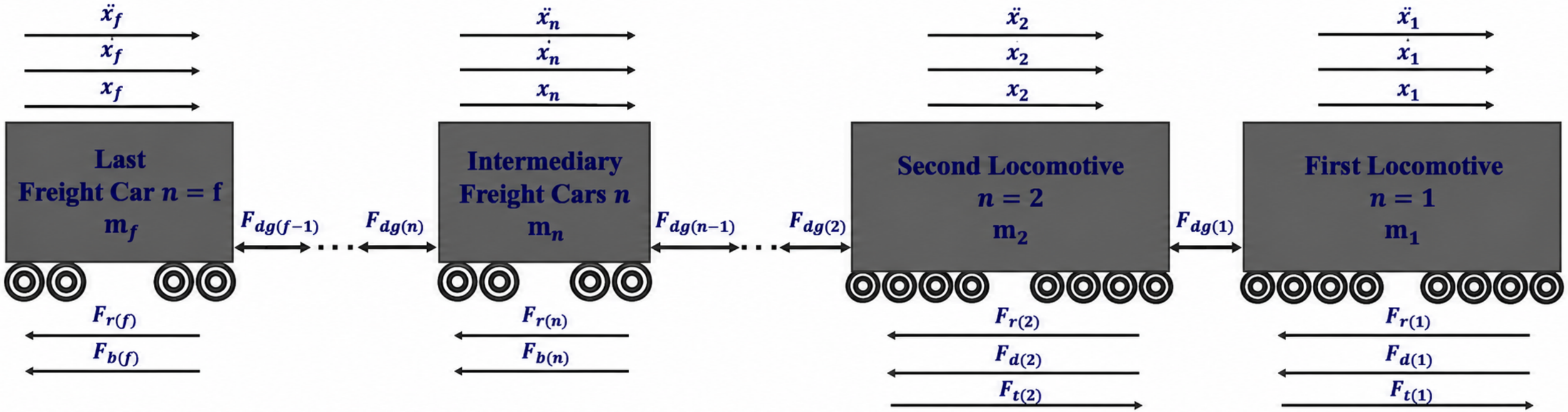

$\ddot{x}_f$
$\dot{x}_f$
x_f
Last
Freight Car $n = f$
m_f
$F_{dg(f-1)}$
$F_{dg(n)}$
$F_{r(f)}$
$F_{b(f)}$
$\ddot{x}_n$
$\dot{x}_n$
x_n
Intermediary
Freight Cars n
m_n
$F_{dg(n-1)}$
$F_{dg(2)}$
$F_{r(n)}$
$F_{b(n)}$
$\ddot{x}_2$
x_2
x_2
Second Locomotive
$n = 2$
m_2
$F_{dg(1)}$
$F_{r(2)}$
$F_{d(2)}$
$F_{t(2)}$
$\ddot{x}_1$
$\dot{x}_1$
x_1
First Locomotive
$n = 1$
m_1
$F_{r(1)}$
$F_{d(1)}$
$F_{t(1)}$